

1. INTRODUCTION

Solar collectors are heat exchangers that use solar radiation to heat a working fluid, usually liquid or air. They can be classified in three groups:

- Flat-plate collectors,
- Evacuated-tube collectors
- Focusing collectors.

- In flat-plate collectors there is no optical concentration of sunlight and they are generally stationary . In addition to this their outlet temperature capability is below 100 °C
- However to reach higher temperatures evacuated-tube collectors and focusing collectors are used. In evacuated-tube collectors they use vacuum to reduce heat lost and to protect the absorber coating from deterioration. By this way they can reach temperatures up to 140 °C and they can collect both direct and diffuse solar radiation
- And focusing collectors, they are not stable and they follow the sun to get direct radiation; they can not utilize diffuse radiation. And they are also capable of producing high temperatures [1].

2. FLAT-PLATE COLLECTORS

A flat plate collector is basically a black surface that is placed at a convenient path of the sun. And a typical flat plate collector is a metal box with a glass or plastic cover (called glazing) on top and a dark-colored absorber plate on the bottom. The sides and bottom of the collector are usually insulated to minimize heat loss.[2]

Figure 2.1 gives examples of flat-plate collectors



Figure 2.1 Flat-plate collectors[3].

Components of a typical flat plate collector:

- **Absorber plate:**

It is usually made of copper, steel or plastic. The surface is covered with a flat black material of high absorptance. If copper or steel is used it is possible to apply a selective coating that maximizes the absorptance of solar energy and minimizes the radiation emitted by plate.

- **Flow passages:**

The flow passages conduct the working fluid through the collector. If the working fluid is a liquid, the flow passage is usually a tube that is attached to or is a part of absorber plate. If the working fluid is air, the flow passage should be below the absorber plate to minimize heat losses.

- **Cover plate:**

To reduce convective and radiative heat losses from the absorber , one or two transparent covers are generally placed above the absorber plate.They usually be made from glass or plastic.

- **Insulation:**

These are some metarials such as fiberglass and they are placed at the back and sides of the collector to rduce heat losses.

- **Enclosure:**

A box that the collector is enclosed in holds the componrnts together, protect them from weather, facilitates installation of the collector on a roof or appropriate frame [1].

Here in figure 2.2 we can see components of flat plate collectors.

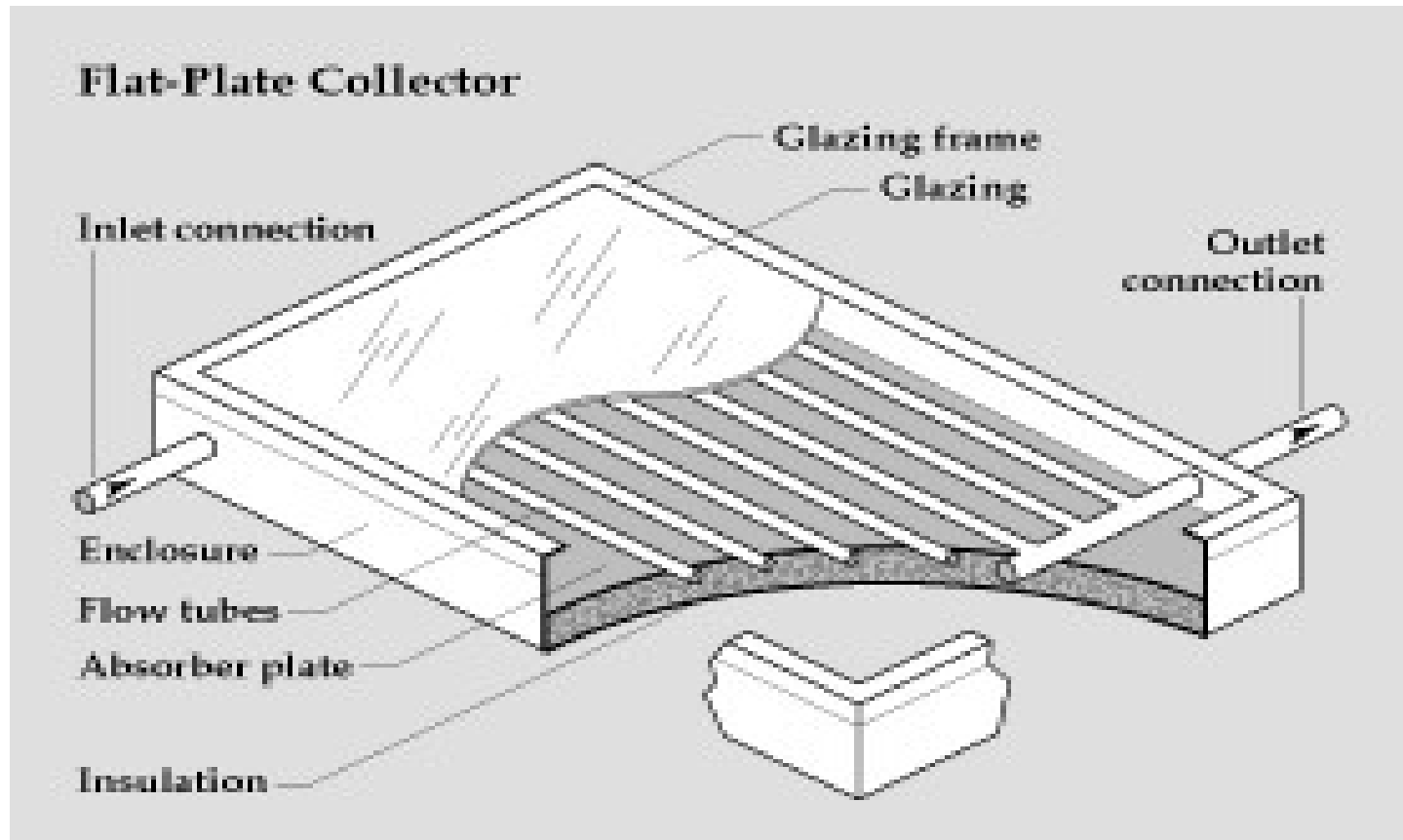
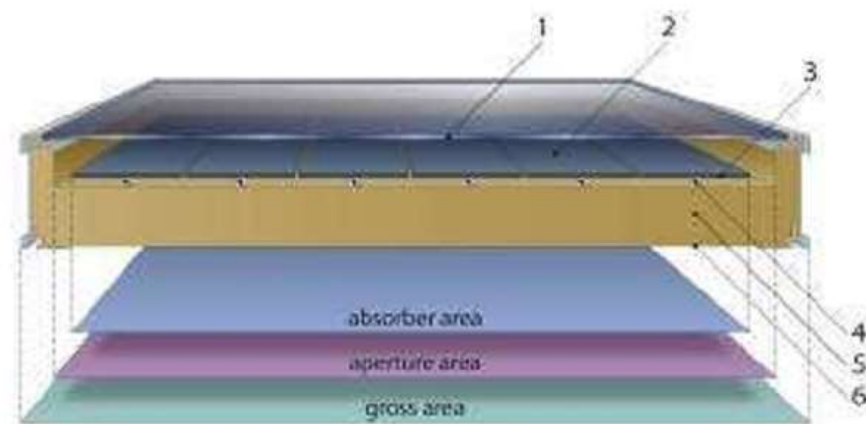
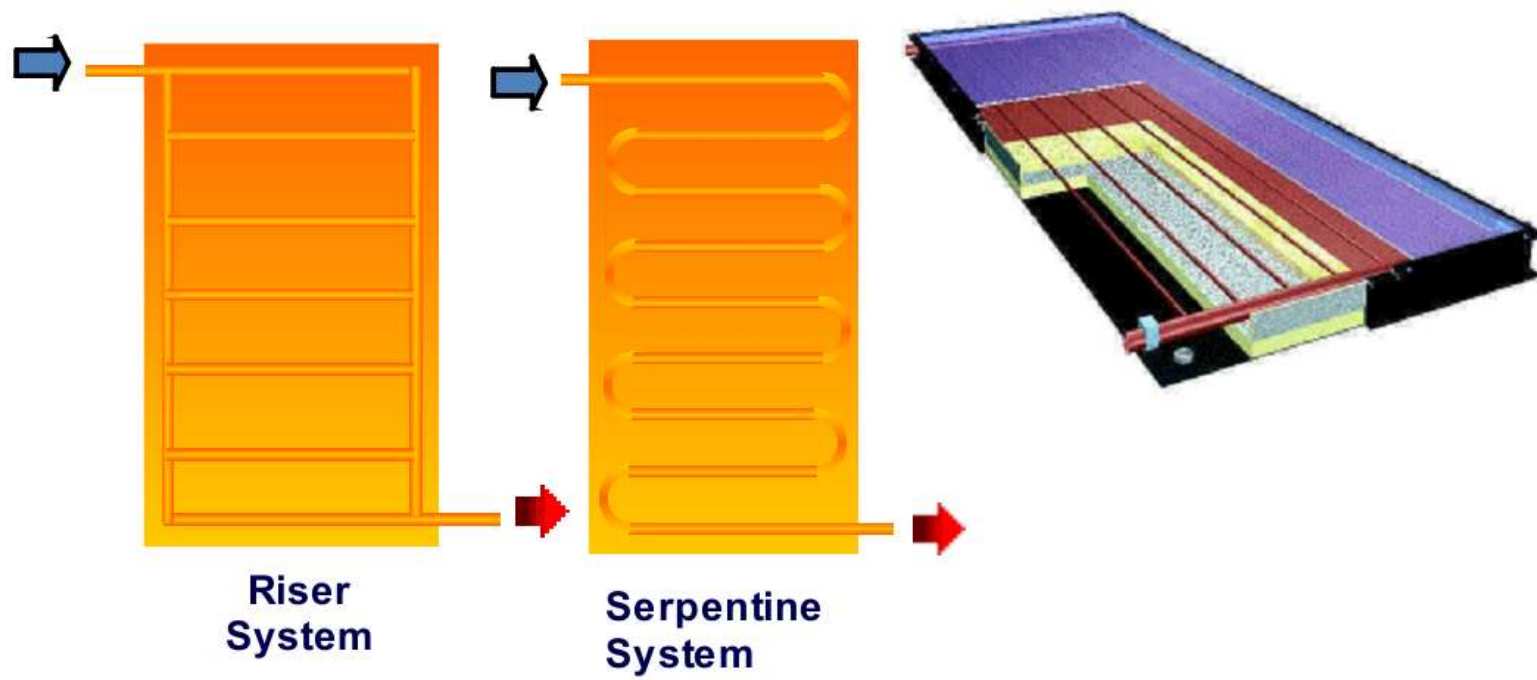


Figure 2.2 Cross section of a basic flat-plate solar collector [4].



A. Absorber plate & Flow passages

Copper, which has high conductivity and is corrosion-resistant, is the material for absorber plates, but because copper is expensive, steel is also widely used. For a copper plate 0.05 cm thick with 1.25-cm tubes spaced 15 cm apart in good thermal contact with the copper, the fin efficiency is better than 97 percent.

The surface of the absorber plate determines how much of the incident solar radiation is absorbed and how much is emitted at a given temperature. Flat black paint which is widely used as a coating has an absorptance of about 95 percent for incident shortwave solar radiation. It is durable and easy to apply [1].

Here a table about matters that absorber plate may be made from

Material	Absorptance (α)	Emittance (ϵ)	Break down temperature (°C)	Comments
Black silicon paint	0.86-0.94	0.83-0.89	350	Slicone binder
Black silicon paint	0.9	0.5		Stable at high temperature
Black copper over copper	0.85-0.9	0.08-0.12	450	Patinates with moisture
Black chorome over nickel	0.92-0.94	0.07-0.12	450	Stable at high temperatures

Table 2.1 Characteristics of absorptive coatings [1].

Here in figure 2.3 we can see absorber plate and flow passages

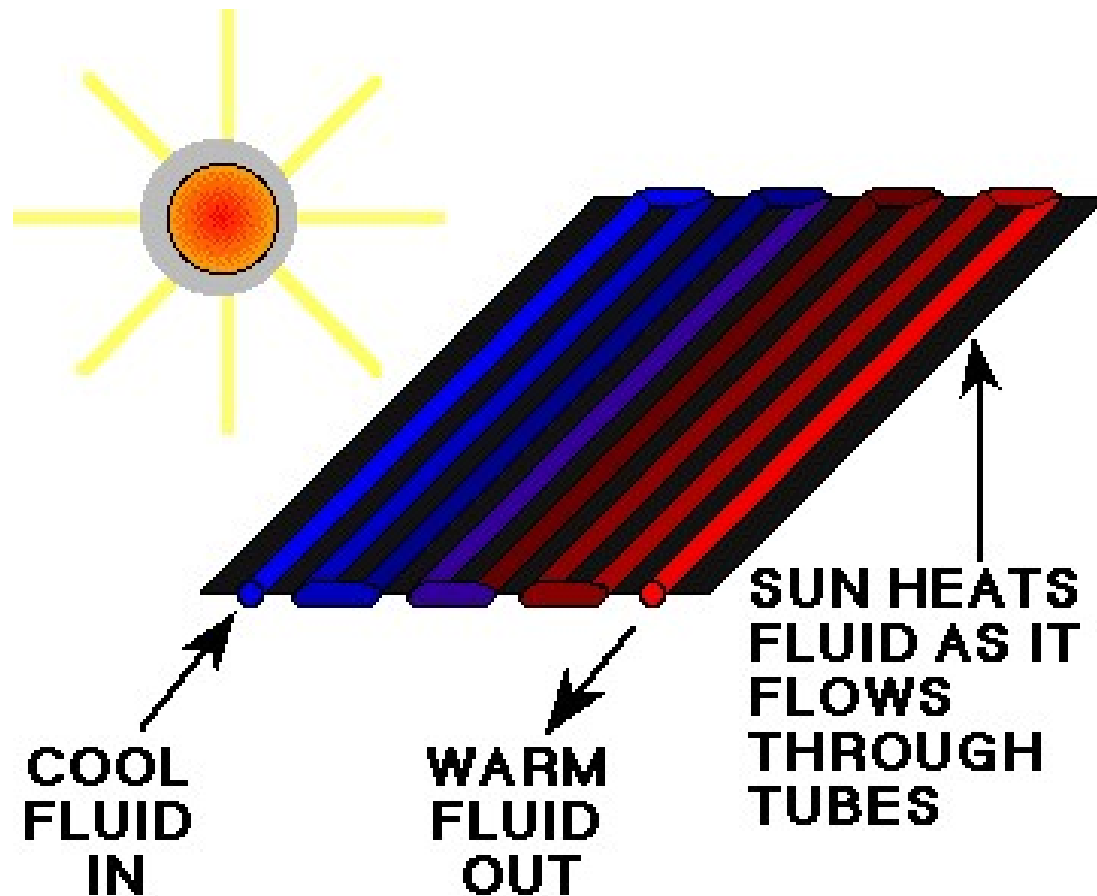


Figure 2.3 Cross section of a absorber plate&flow passages of a flat plate collector [4].

B. Cover plates

A cover plate for a collector should have a high transmittance for solar radiation and should not deteriorate with time. The material most commonly used is glass. A 0.32-cm thick sheet of window glass (iron content, 0.12 percent) transmits 85 percent of solar energy at normal incidence. And all glass is practically opaque to long-wavelength radiation emitted by the absorber plate.

Some plastic materials can be used for collector glazing. They are cheaper and lighter than glass and, because they can be used in very thin sheets, they often have higher transmittance. However, they are not as durable as glass and they often degrade with exposure to ultraviolet radiation or high temperatures [1].

Here a table about matters that cover plate may be made from

Test	Polyvinly floride	Polyethylene terephthatalet or polyster	Polycarbonate	Fiberglass rein forced plastics
Solar Transmission, %	92-94	85	82-89	77-90
Maximu operating temperature ° C	110	100	120-135	95
Thermal Expansion Coefficient	43	27	68	32-40
Thickness, mm	0.1	0.025	3.2	1.0
Length of life, years	In 5 years 95% retains	4		7-20

Table 2.2 Charactericts of cover plate materials [1].

Here in figure 2.4 we can see cover part.

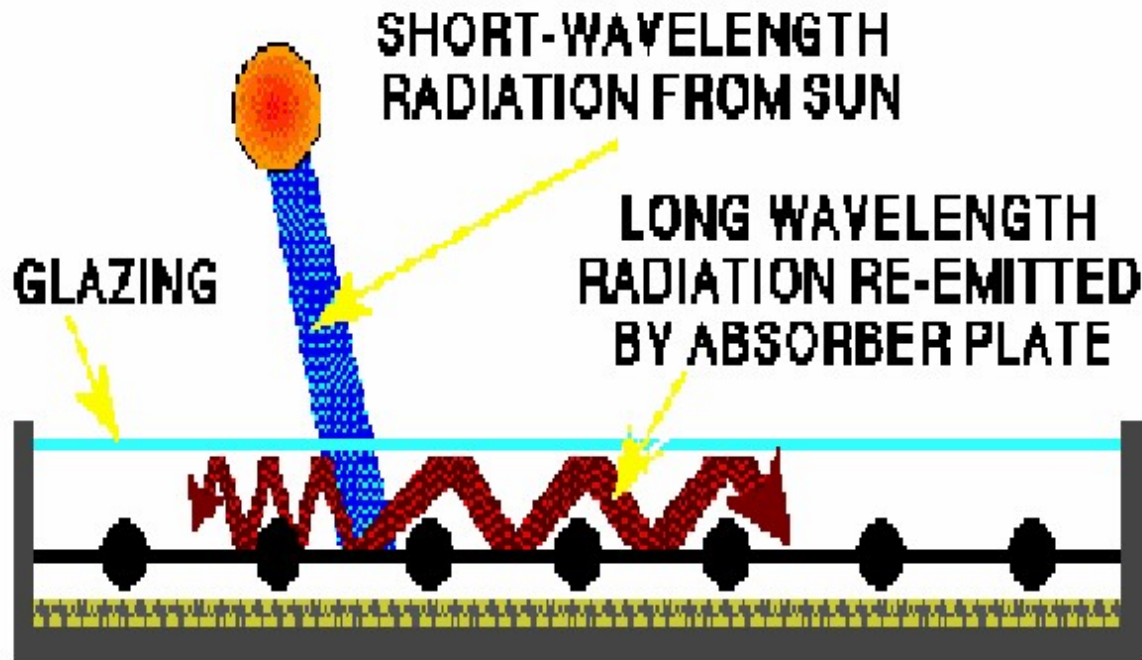


Figure 2.4 Cross section of a cover part of a flat-plate collector [4].

C. Enclosure / Insulation

The collector enclosure is usually made from steel, aluminium or fiber glass. And order to prevent heat from escaping through the back of the collector, a layer of insulation is placed behind the absorber plate [1].

Material	Density Kg/m³	Thermal conductivity at 95 °C (W/mK)	Temperature limits °C
Fiber glass with organic binder	11	0.059	175
“	16	0.050	175
“	24	0.045	175
“	48	0.43	175

Table 2.3 Characteristics of insulation materials [1].

Here in figure 2.5 we can see insulation part.



Figure 2.5 Cross Section of an Insulation Part of a Flat-Plate Collector [4].

3. PROPER ORIENTATION and ANGLE of SOLAR COLLECTOR

Flat plate collectors are divided in three main groups according to how they are oriented:

- Flat-plate collectors facing south at fixed tilt
- One-axis tracking flat-plate collectors with axis oriented north-south
- Two-axis tracking flat-plate collectors

A. Flat-plate collectors facing south at fixed tilt:

To optimize performance in the winter, the collector can be tilted 15° greater than the latitude; to optimize performance in the summer, the collector can be tilted 15° less than the latitude [5]. Figure 3.1 show how the collector is tilted.

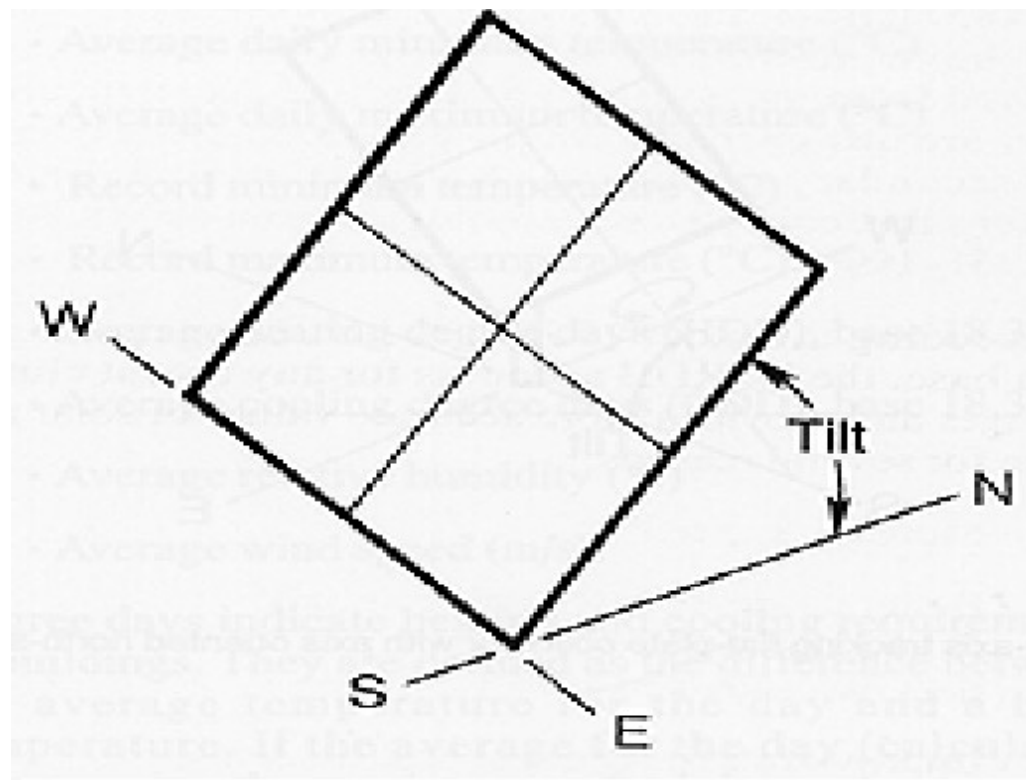


Figure 3.1 Flat-plate collector at fixed tilt [5].

Energy diagram of typical flat plate collector is shown in figure 5.1. % 92 of the total sunshine reaches to the copper absorber. % 8 of the total sunshine is reflected from glass. % 5 of the sunshine is emitted from the panel, %12 is lost through convection and conduction.

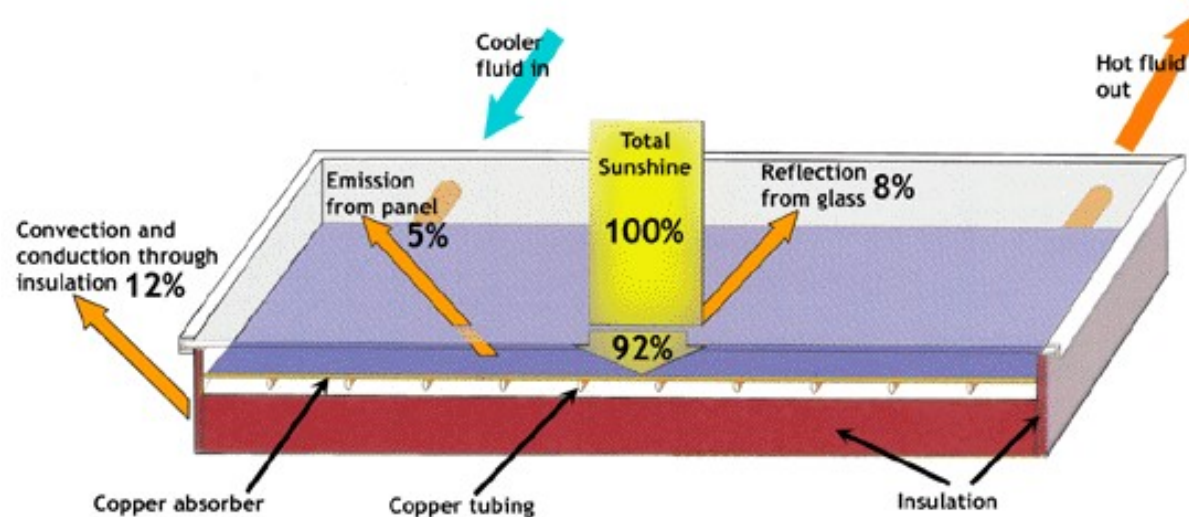


Figure 5.1 Energy diagram for typical flat plate collector [3]

6) APPLICATIONS

Flat plate collectors are used for both;

A) Domestic applications

B) Commercial applications

A) Domestic applications

Flat plate collectors mainly used in residential buildings where the demand for hot water has a large impact on energy bills. This generally means a situation with a large family, or a situation in which the hot water demand is excessive due to frequent laundry washing [2].

For instance, a family of 4 members consumes on an average 100 litre of hot water a day at 60 °C. Hot water of 100 litre capacity at 60 °C approximate can be delivered by a single collector system of 2 m² area. The solar water heating systems are generally provided with auxiliary backup in the insulated hot storage tank for the rainy and heavily overcast cloudy days [7].

Here we can see solar flat-plate collectors used for heating buildings.



Figure 6.1 Flat plate collectors used for heating buildings [8].

B) Commercial applications

Commercial applications include laundromats, car washes, military laundry facilities and eating establishments. Solar water heating systems are most likely to be cost effective for facilities with water heating systems that are expensive to operate, or with operations such as laundries or kitchens that require large quantities of hot water.

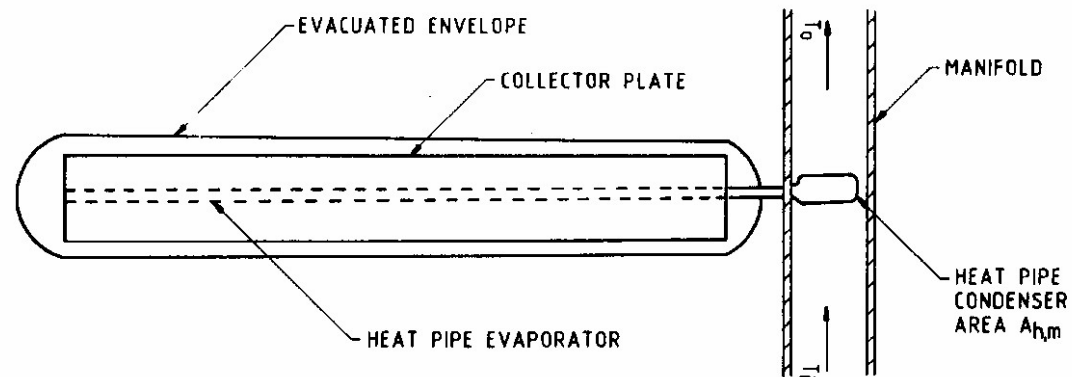
And unglazed liquid collectors are commonly used to heat water for swimming pools. Because these collectors need not withstand high temperatures, they can use less expensive materials such as plastic or rubber. They also do not require freeze-proofing because swimming pools are generally used only in warm weather or can be drained easily during cold weather [2].

Here we can see solar flat-plate collectors used for heating swimming pools.

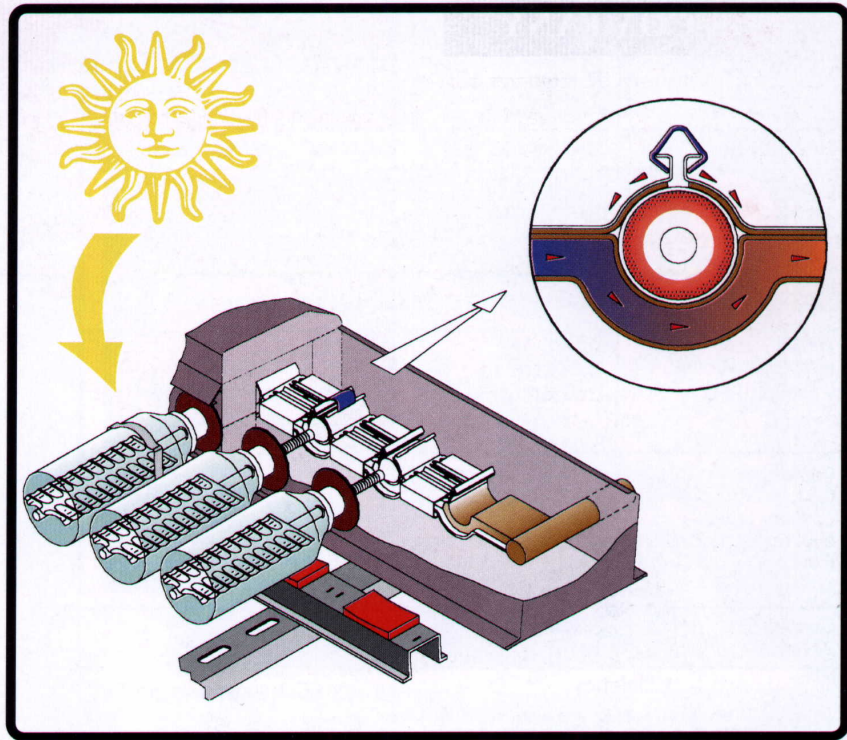


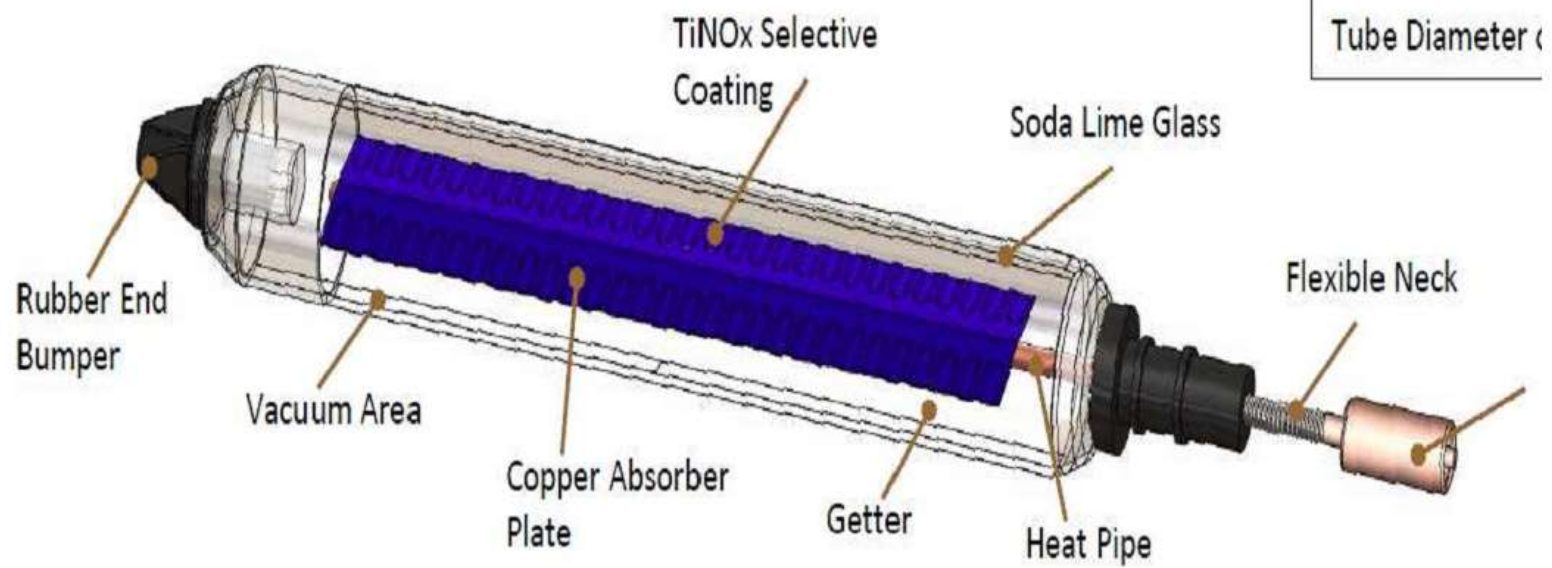
Figure 6.2 Flat-plate collectors used for heating swimming pools [9].

Schematic diagram of an evacuated tube collector



Evacuated tube collectors





Vacuum Tubes

Advantages

- Higher solar yield than flat plate collectors with the same absorber area.
- 30% more effective
- Little thermal loss (only through some radiation)
- As the tube is sealed, dirt or moisture cannot get into the collector
- Works in cold, windy and humid conditions
- Individual tubes can be rotated to optimise ideal orientation
- High temperatures can be achieved
- Some people like the aesthetics of tubes

Disadvantages

- Cost !
- Not easily integrated into the fabric of the building (e.g. roof or facade integrated)
- Some people do not like the aesthetics of the tubes

Flat Plates

Advantages

- Lower cost
- Can be more easily integrated with the fabric of the building (roof integrated / facade integrated)

Disadvantages

- Lower performance through conduction, convection and radiation therefore you need more surface area of collector
- Higher wind load requirements (they act like a sail and catch the wind)
- Glass and absorber can get dirty and effect performance
- Needs more maintenance
- Higher volume content equated to larger component sizes. i.e. expansion vessel
- Isn't as efficient in lower light

7) CONCLUSION

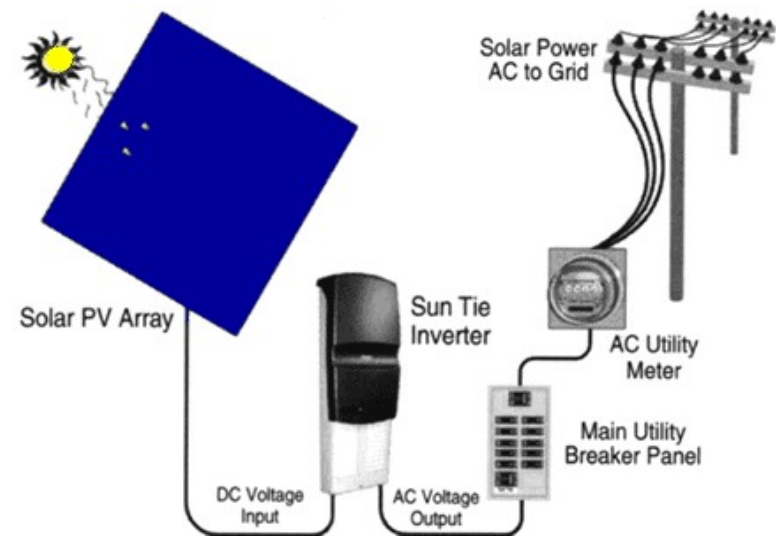
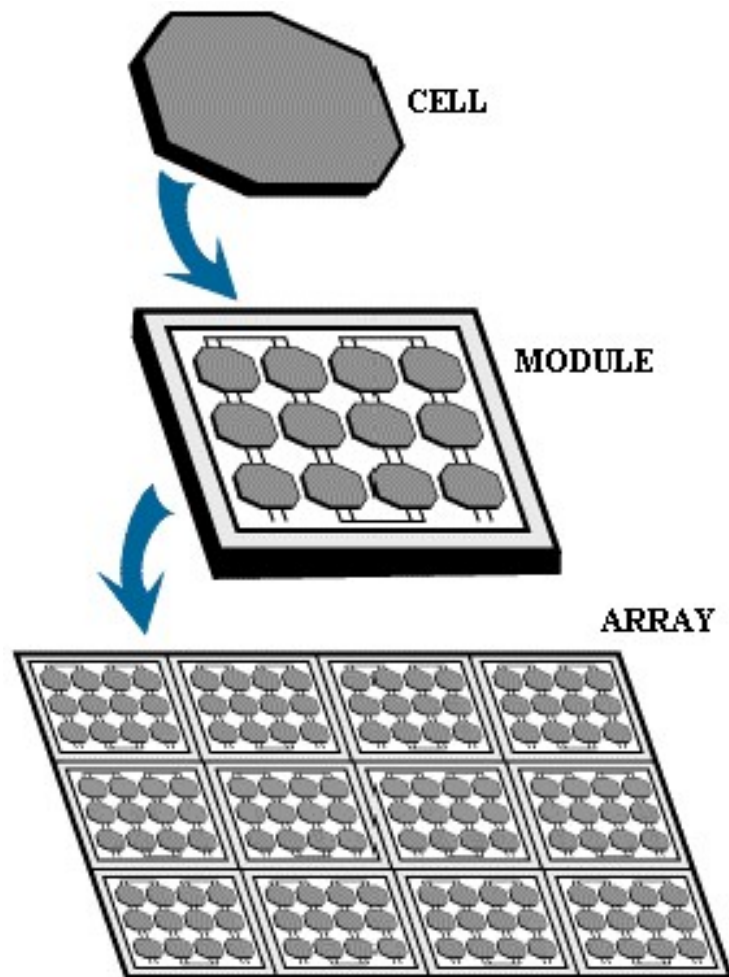
Flat-plate collectors which are used for water heating, are long lasting, and also in long term they are cheaper than other water heating systems. However, they requires

large areas if high energy output is a requirement.

Than solar energy is free if we do not include the initial cost for installation and the maintenance.

Finally; besides these we should remember by using solar energy we can protect nature.

PV CELL TECHNOLOGY



PV Technology Classification

Silicon Crystalline Technology

- Mono Crystalline PV Cells
- Multi Crystalline PV Cells

Thin Film Technology

- Amorphous Silicon PV Cells
- Poly Crystalline PV Cells
(Non-Silicon based)

Silicon Crystalline Technology

- Currently makes up 86% of PV market
- Very stable with module efficiencies 10-16%

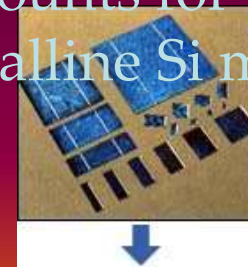
Mono crystalline PV Cells

- Made using saw-cut from single cylindrical crystal of Si
- Operating efficiency up to 15%



Multi Crystalline PV Cells

- Cast from ingot of melted and recrystallised silicon
- Cell efficiency ~12%
- Accounts for 90% of crystalline Si market



Multicrystal silicon solar cells



Thin Film Technology

- Silicon deposited in a continuous on a base material such as glass, metal or polymers
- Thin-film crystalline solar cell consists of layers about $10\mu\text{m}$ thick compared with $200\text{--}300\mu\text{m}$ layers for crystalline silicon cells

PROS

- Low cost substrate and fabrication process

CONS

- Not very stable



Amorphous Silicon PV Cells

- The most advanced of thin film technologies
- Operating efficiency ~6%
- Makes up about 13% of PV market

PROS

- Mature manufacturing technologies available

CONS

- Initial 20-40% loss in efficiency



Poly Crystalline PV Cells

Copper Indium Diselenide

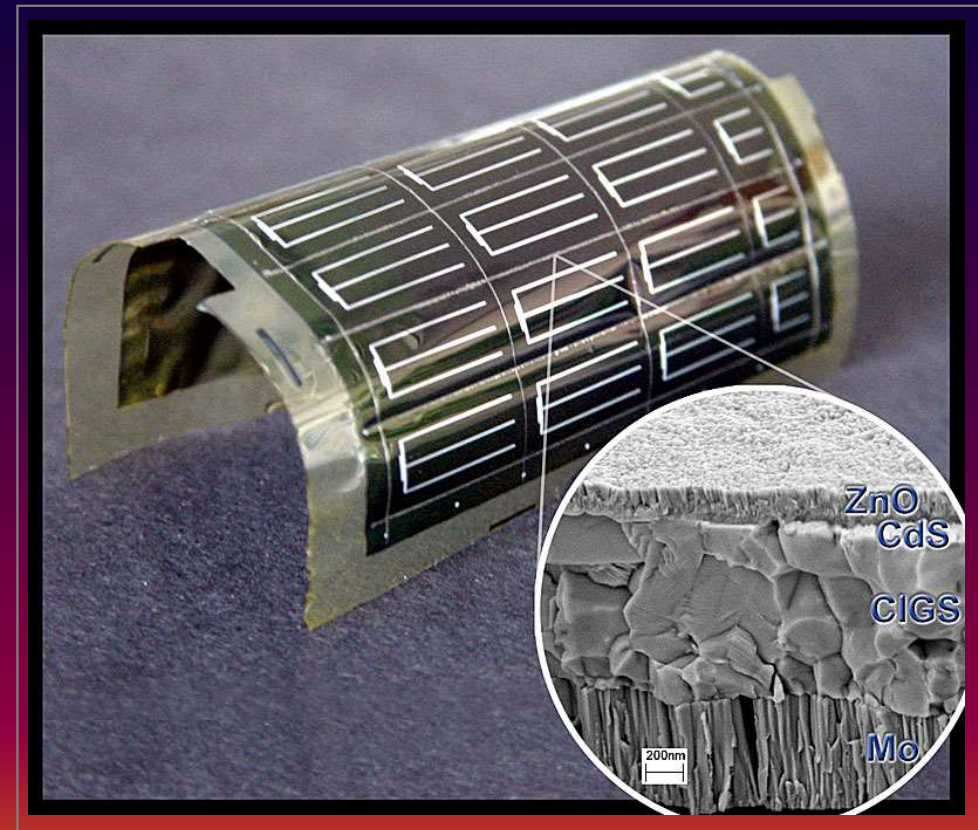
- CIS with band gap 1eV, high absorption coefficient 10^5cm^{-1}
- High efficiency levels

PROS

- 18% laboratory efficiency
- >11% module efficiency

CONS

- Immature manufacturing process
- Slow vacuum process



Poly Crystalline PV Cells

Non – Silicon Based Technology

Cadmium Telluride (CdTe)

- Unlike most other II/IV material CdTe exhibits direct band gap of 1.4eV and high absorption coefficient

PROS

- 16% laboratory efficiency
- 6-9% module efficiency

CONS

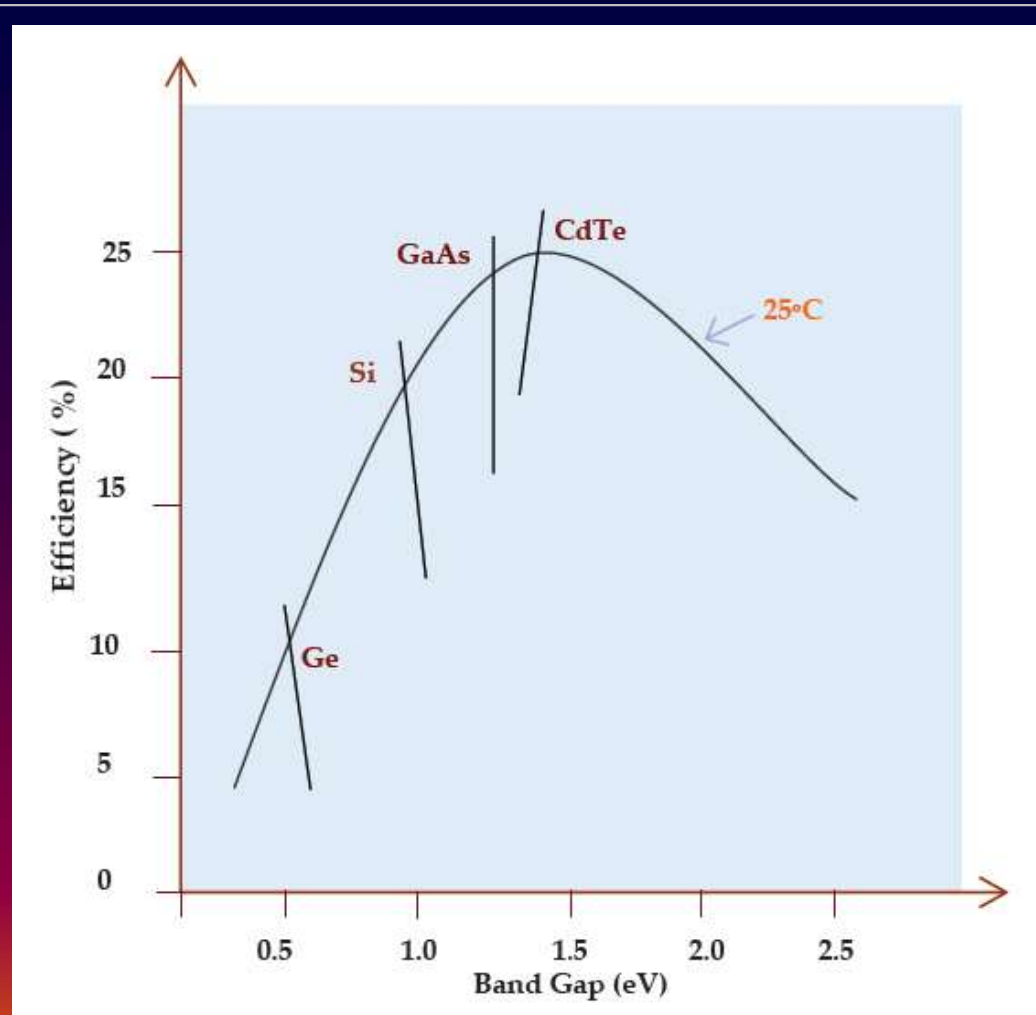
- Immature manufacturing process



Semiconductor Material Efficiencies

- Germanium
- Silicon
- Gallium Arsenide (GaAs)
- Cadmium-Tellurium

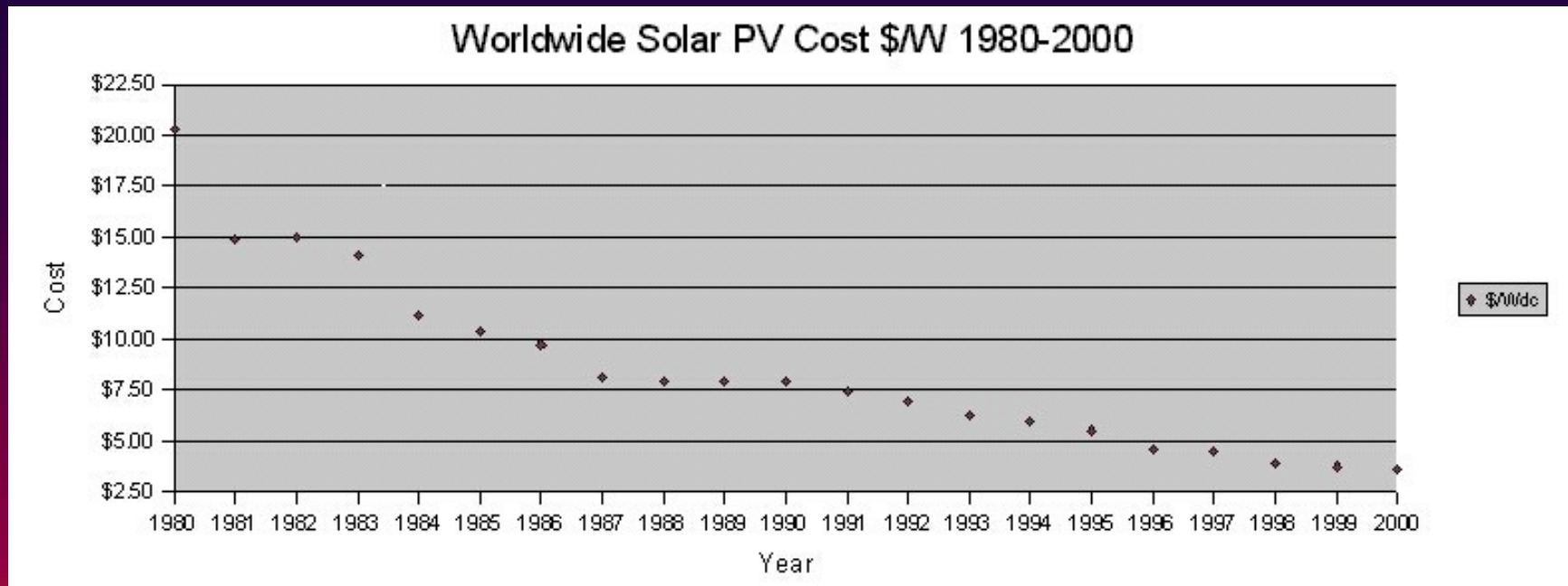
Semiconductor Material Efficiencies



PV Economics

- PV unit : Price per peak watt (Wp)
(Peak watt is the amount of power output a PV module produces at Standard Test Conditions (STC) of a module operating temperature of 25°C in full noontime sunshine (irradiance) of 1,000 Watts per square meter)
- A typical 1kWp System produces approximately 1600-2000 kWh energy in India and Australia
- A typical 2000 watt peak (2KWp) solar energy system costing \$8000 (including installation) will correspond to a price of \$4/Wp

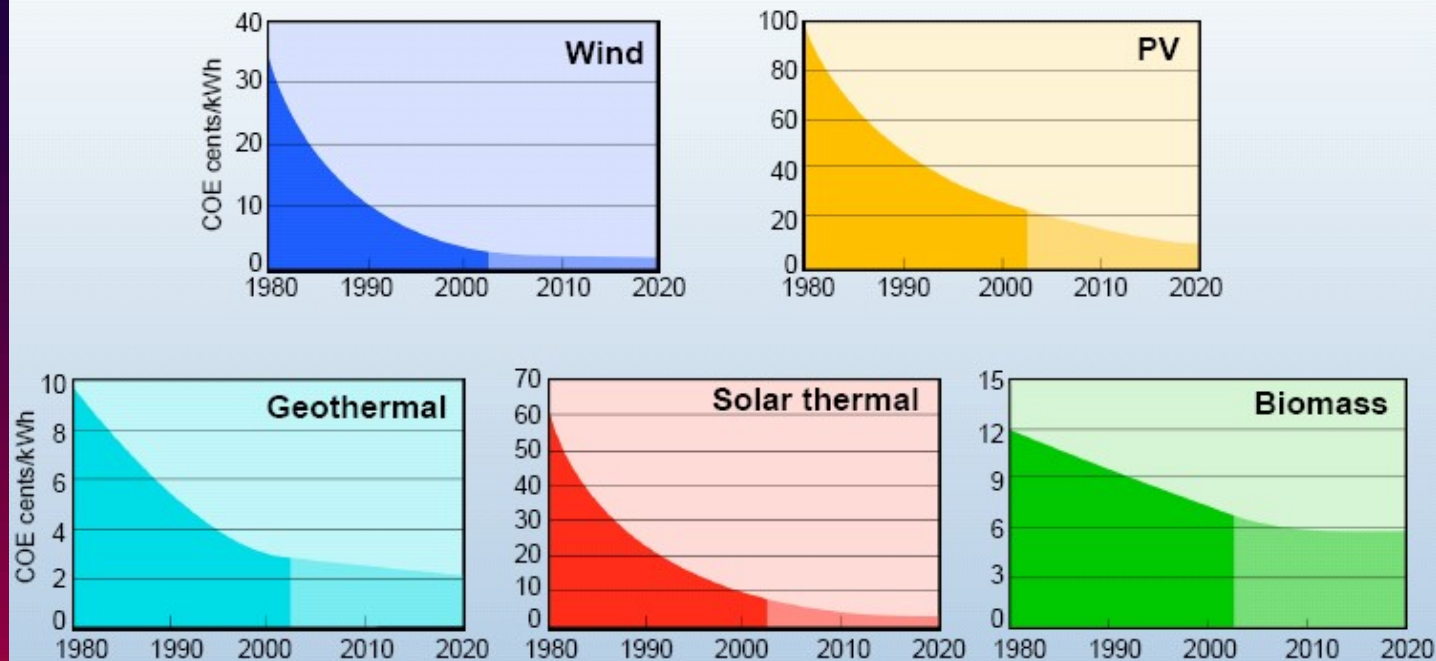
Solar PV Costs 1980-2000



There has been almost six fold decline in price per peak watt of PV module from 1980 to year 2000

Renewable Energy Cost Trends

Levelized cents/kWh in constant \$2000¹



Source: NREL Energy Analysis Office (www.nrel.gov/analysis/docs/cost_curves_2002.ppt)

¹These graphs are reflections of historical cost trends NOT precise annual historical data.

Updated: October 2002

Solar electricity prices are today, around 30 cents/kWh, but still 2-5 times average Residential electricity tariffs

Applications @ PV

- **Water Pumping:** PV powered pumping systems are excellent ,simple ,reliable – life 20 yrs
- **Commercial Lighting:** PV powered lighting systems are reliable and low cost alternative. Security, billboard sign, area, and outdoor lighting are all viable applications for PV
- **Consumer electronics:** Solar powered watches, calculators, and cameras are all everyday applications for PV technologies.
- **Telecommunications**
- **Residential Power:** A residence located more than a mile from the electric grid can install a PV system more inexpensively than extending the electric grid
(Over 500,000 homes worldwide use PV power as their only source of electricity)